CHAPTER 17 SECOND-ORDER DIFFERENTIAL EQUATIONS

17.1 Second-Order Linear Equations

- A **second-order linear** differential equation has the form

$$P(x)\frac{d^2y}{dx^2} + Q(x)\frac{dy}{dx} + R(x)y = G(x)$$

where P, Q, R, and G are continuous functions.

  - **Homogeneous** Linear Equation $\qquad G(x) = 0$ for all x

E.g. $\qquad \dfrac{d^2y}{dx^2} + 2\dfrac{dy}{dx} + y = 0$

right HS is a function of x

  - **Nonhomogeneous** Linear Equation $\qquad G(x) \neq 0$ for some x

E.g. $\qquad x^2\dfrac{d^2y}{dx^2} + 3\sin(x)\dfrac{dy}{dx} + 2y = 2x$

c only test this!

- If y_1, y_2, …, y_n are functions,

$$c_1y_1 + c_2y_2 + \ldots + c_ny_n$$

is called a **linear combination** of y_1, y_2, …, y_n.

- We say that the functions y_1, y_2, …, y_n are **linearly independent** if and only if

$$c_1y_1 + c_2y_2 + \cdots + c_ny_n = 0$$

implies that

$$c_1 = c_2 = \cdots = c_n = 0.$$

- **Theorem**

If $y_1(x)$ and $y_2(x)$ are both solutions of the linear homogeneous equation

$$P(x)\frac{d^2y}{dx^2} + Q(x)\frac{dy}{dx} + R(x)y = 0$$

then the linear combination

$$y(x) = c_1y_1(x) + c_2y_2(x)$$

is also a solution.

eg $y_1 = e^x$

$y_2 = 4e^x$

- **Theorem**

If y_1 and y_2 are linearly independent solutions of the linear homogeneous equation

$$P(x)\frac{d^2y}{dx^2} + Q(x)\frac{dy}{dx} + R(x)y = 0$$

on an interval I, and $P(x)$ is never 0 on I, then the general solution is given by

$$y(x) = c_1y_1(x) + c_2y_2(x)$$

where c_1 and c_2 are arbitrary constants.

- This theorem is very useful because it says that if we know *two* particular linearly independent solutions, then we know *every* solution.

- **Second-Order Linear Homogeneous Equation with Constant Coefficients**

- The differential equation of this type takes the form

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

where a, b, and c are constants and $a \neq 0$.

Summary: Solutions of $ay'' + by' + c = 0$

Roots of $ar^2 + br + c = 0$	General solution
r_1, r_2 real and distinct	$y = c_1e^{r_1x} + c_2e^{r_2x}$
$r_1 = r_2 = r$	$y = c_1e^{rx} + c_2xe^{rx}$
r_1, r_2 complex: $\alpha \pm i\beta$	$y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$

- **Characteristic equation** of the differential equation

$$ar^2 + br + c = 0$$

- **Case 1**: $b^2 - 4ac > 0$

- **Case 2**: $b^2 - 4ac = 0$

- **Case 3**: $b^2 - 4ac < 0$

Example 9.1. Solve the differential equation

$$y'' + y' - 6y = 0$$

let $y = e^{rx}$

$y' = r \cdot e^{rx}$

$y'' = r^2 \cdot e^{rx}$

$r^2 e^{rx} + r e^{rx} - 6e^{rx} = 0$

$e^{rx}(r^2 + r - 6) = 0$

$e^{rx} = 0$ or $r^2 + r - 6 = 0$

impossible

$(r+3)(r-2) = 0$

$r = -3$ or $r = 2$

$y_1 = e^{-3x}$

$y_2 = e^{2x}$

$y = C_1 e^{-3x} + C_2 e^{2x}$

Example 9.2. Solve the differential equation

$$3\frac{d^2y}{dt^2} + \frac{dy}{dt} - y = 0$$



Summary: Solutions of $ay'' + by' + c = 0$

Roots of $ar^2 + br + c = 0$	General solution
r_1, r_2 real and distinct	$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$
$r_1 = r_2 = r$	$y = c_1 e^{rx} + c_2 x e^{rx}$
r_1, r_2 complex: $\alpha \pm i\beta$	$y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$

Example 9.3. Solve the differential equation

$$4y'' + 12y' + 9y = 0.$$

特徵方程

let $y = e^{rx}$

$y' = re^{rx}$

$y'' = r^2 e^{rx}$

$4r^2 e^{rx} + 12re^{rx} + 9e^{rx} = 0$

$e^{rx}(4r^2 + 12r + 9) = 0$

$e^{rx} = 0$ impossible

$4r^2 + 12r + 9 = 0$

$(2r+3)^2 = 0$

$r = -\frac{3}{2}$

$\therefore y =$

Example 9.4. Solve the differential equation

$$y'' - 6y' + 13y = 0.$$

Example 9.5. Solve the initial-value problem

$$y'' + y' - 6y = 0, \quad y(0) = 1, \quad y'(0) = 0$$

let $y = e^{rx}$

$y' = re^{rx}$

$y'' = r^2 e^{rx}$

$r^2 e^{rx} + re^{rx} - 6e^{rx} = 0$

$e^{rx}(r^2 + r - 6) = 0$

$e^{rx} = 0$ or $r^2 + r - 6 = 0$

$(r+3)(r-2) = 0$

$r = -3$ or $r = 2$

General Solution $\Rightarrow$ $y = c_1 e^{-3x} + c_2 e^{2x}$

$y' = -3c_1 e^{-3x} + 2c_2 e^{2x}$

Given $y(0) = 1$: $1 = c_1 + c_2$ --- (1)

Given $y'(0) = 0$: $0 = -3c_1 + 2c_2$ --- (2)

(1) $\times 2$ - (2)

$2 - 0 = -5c_1$

$c_1 = -\frac{2}{5}$

from (1)

$c_2 = \frac{7}{5}$

$\therefore y = -\frac{2}{5} e^{-3x} + \frac{14}{5} e^{2x}$

Particular solution $\Rightarrow$

Example 9.6. Solve the initial-value problem

$$y'' + y = 0, \quad y(0) = 2, \quad y'(0) = 3$$

Example 9.7. Solve the boundary-value problem

$y = e^{-x} + e^{-2x}$

$$y'' + 2y' + y = 0, \quad y(0) = 1, \quad y(1) = 3$$

Exercises 17.1 1, 5, 9, 13, 17, 19, 21, 25, 29, 30 ($y = 4e^{x/2} - 2xe^{x/2}$), 31, 33.

17.2 Nonhomogeneous Linear Equations

- **Theorem**

The general solution of the nonhomogeneous equation

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = G(x)$$

is of the form

$$y(x) = y_p(x) + y_c(x)$$

where $y_p(x)$ is a particular solution of the equation, and $y_c(x)$ is the general solution of the complementary equation

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

- There are two methods to find $y_p(x)$:
  1. **Method of Undetermined Coefficients**: straigthforward but works only for a restricted class of functions G.
  2. **Method of Variation of Parameters**: works for every function G but is usually more difficult to apply in practice.

- **The Method of Undetermined Coefficients**

Example 9.8. Solve the differential equation

$$y'' + y' - 2y = x^2$$

① y_c ② Guess $y_p(x) = Ax^2 + Bx + C$

let $y_c = e^{rx}$ such that $y'' + y' - 2y_c = 0$

$y_c' = re^{rx}$

$y_c'' = r^2 e^{rx}$

$e^{rx}(r + r^2 - 2) = 0$

$e^{rx} = 0$ impossible or $r + r^2 - 2 = 0$

$(r+2)(r-1) = 0$

$r = -2$ or $r = 1$

$y_c = c_1 e^{-2x} + c_2 e^{x}$

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① y_p

Assume $y_p = Ax^2 + Bx + C$

let $y_p = Ax^2 + Bx + C$

$y_p' = 2Ax + B$

$y_p'' = 2A$

$y'' + y' - 2y = x^2$

$2A + 2Ax + B - 2(Ax^2 + Bx + C) = x^2$

$-2Ax^2 + (2A - 2B)x + (2A + B - 2C) = x^2$

Compare left and right

$-2A = 1$

$2A - 2B = 0$ $\Rightarrow$ $A = -\frac{1}{2}$, $B = -\frac{1}{2}$, $C = -\frac{3}{4}$

$2A + B - 2C = 0$

$y_p = -\frac{1}{2}x^2 - \frac{1}{2}x - \frac{3}{4}$

② $y = y_c + y_p$

$= c_1 e^{-2x} + c_2 e^{x} - \frac{1}{2}x^2 - \frac{1}{2}x - \frac{3}{4}$

undetermined index

Example 9.9. Solve the differential equation

$$y'' + 4y = e^{3x}$$

$y(p) = Ae^{3x}$

Example 9.10. Solve the differential equation

$$y'' + y' - 2y = \sin x$$

$y_p = A\sin x + B\cos x$

Example 9.11. Solve the differential equation

$$y'' - 4y = xe^x + \cos 2x$$

$Ax+B$ ↑ Ae^x ← xe^x (δ)

$y_p = (Ax+B)e^x + C\cos 2x + D\sin 2x$

Example 9.12. Solve the differential equation

$$y'' + y = \sin x \quad \Rightarrow \quad A\sin x + B\cos x$$

let $y_c = e^{rx}$ such that $y'' + y = 0$

$$e^{rx}(1 + r^2) = 0$$

$e^{rx} = 0$ $\qquad 1 + r^2 = 0$

$r^2 = -1$

$r = \pm i$

Summary: Solutions of $ay'' + by' + c = 0$

Roots of $ar^2 + br + c = 0$	General solution
r_1, r_2 real and distinct	$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$
$r_1 = r_2 = r$	$y = c_1 e^{rx} + c_2 x e^{rx}$
r_1, r_2 complex: $\alpha \pm i\beta$	$y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$

$\therefore r = 0 \pm i \cdot 1$

$\alpha = 0, \quad \beta = 1$

$\therefore y = C_1 \cos x + C_2 \sin x$

let $y_t = x(A\sin x + B\cos x)$

$y'_t = A\sin x + B\cos x + x(A\cos x - B\sin x)$

$y''_t = A\cos x - B\sin x + A\cos x - B\sin x + x(-A\sin x - B\cos x)$

$\Rightarrow \quad y'' + y = \sin x$

$2A\cos x - 2B\sin x = \sin x$

$A = 0, \quad B = -\frac{1}{2}$

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to some coefficients. Then we substitute this solution into the equation to determine the coefficients. Table 1.1 shows some specific examples of $g(x)$ along with the corresponding form of the particular solution.

$g(x)$	Form of y_p
1	A
$2x+1$	$Ax+B$
$2x^2+3x+4$	Ax^2+Bx+C
$4x^3+5x^2-1$	Ax^3+Bx^2+Cx+E
$\sin 3x$	$A\cos 3x+B\sin 3x$
$\cos 3x$	$A\cos 3x+B\sin 3x$
e^{9x}	Ae^{9x}
$(2x+1)e^{9x}$	$(Ax+B)e^{9x}$
x^2e^{2x}	$(Ax^2+Bx+C)e^{2x}$
$e^{3x}\sin 4x$	$Ae^{3x}\cos 4x+Be^{3x}\sin 4x$
$2x^2\sin 4x$	$(Ax^2+Bx+C)\cos 4x+(Ex^2+Fx+G)\sin 4x$
$xe^{2x}\cos 3x$	$(Ax+B)e^{2x}\cos 3x+(Cx+E)e^{2x}\sin 3x$

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- **Summary**

- If $G(x) = e^{kx}P(x)$, where P is a polynomial of degree n, then try

$$y_p(x) = e^{kx}Q(x)$$

where Q is a polynomial of degree n.

- If $G(x) = e^{kx}P(x)\cos mx$ or $G(x) = e^{kx}P(x)\sin mx$, where P is a polynomial of degree n, then try

$$y_p(x) = e^{kx}Q(x)\cos mx + e^{kx}R(x)\sin mx$$

where Q and R are polynomials of degree n.

- If any term of y_p is a solution of the complementary equation, multiply y_p by an appropriate power of x.

Example 9.13. Determine the form of the trial solution for the differential equation

$$y'' - 4y' + 13y = e^{2x}\cos 3x$$

$y_p = [A\cos 3x + B\sin 3x)e^{2x}$

let $y_c = e^{rx}$ such that $y'' - 4y' + 13y = e^{2x}\cos 3x$

$y_c' = r\cdot e^{rx}$

$y_c'' = r^2 e^{rx}$

$e^{rx}(r^2 - 4r + 13) = 0$

$e^{rx} = 0$ (impossible) or $r^2 - 4r + 13 = 0$

$r = 2 \pm 3i$

$y_c = e^{2x}(c_1\cos 3x + c_2\sin 3x)$

$y'' - 4y' + 13y = e^{2x}\cos 3x$

$y_p = [(A\cos 3x + B\sin 3x)e^{2x}]x$?

- **The Method of Variation of Parameters**

- We look for particular solution of the form

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$$

where $y_1(x)$ and $y_2(x)$ are linearly independent solutions of the complementary equation, $u_1(x)$ and $u_2(x)$ are arbitrary functions.

Example 9.14. Solve the differential equation

$$\frac{d^2y}{dx^2} + y = \tan x, \quad 0 < x < \frac{\pi}{2}$$

Exercises 17.2 1, 3, 5, 7, 13, 17, 21, 23, 25.